Zhengjie Miao

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Education

2017-2022 **Ph.D. in Computer Science**, *Duke University*, Durham, NC.
- Dissertation: Simplifying Human-in-the-loop Data Science Pipeline: Explanations, Debugging, and Data Preparation
- Advisor: Sudeepa Roy

2015–2016 **M.S. in Computer Science**, *Columbia University*, New York, NY.
- Advisor: Eugene Wu

2011–2015 **B.S. in Computer Science and Technology**, *Peking University*, Beijing, China.

Professional Experience

10/2023-Present **Assistant Professor**, *Simon Fraser University, School of Computing Science*, Burnaby, BC.

9/2022-10/2023 **Research Scientist**, *Megagon Labs*, Mountain View, CA.

5/2021-8/2021 **Research Intern**, *Microsoft Research (Data Systems Group)*, supervised by Dr. Yeye He, Durham, NC (remote).

5/2020-8/2020 **Research Intern**, *Megagon Labs*, supervised by Dr. Yuliang Li and Dr. Wang-Chiew Tan, Durham, NC (remote).

5/2019-8/2019 **Research Intern**, *Megagon Labs*, supervised by Dr. Yuliang Li and Dr. Wang-Chiew Tan, Mountain View, CA.

Awards

2022 Outstanding Ph.D. Dissertation, *Department of Computer Science, Duke University*

2020 Microsoft Research PhD Fellowship Finalist

2019 Outstanding Ph.D. Research Initiation Project Award, *Department of Computer Science, Duke University*

2019 VLDB Travel Grant

2018, 2019 ACM SIGMOD Travel Award

2013 The May Fourth Scholarship, *Peking University*

Research Full Papers

VLDB 2026 **Benchmarking the Full Pipeline of Materialized-View-Based Query Rewriting**, *Xinjie Hu and Zhengjie Miao*.
Proceedings of the VLDB Endowment (PVLDB), Vol. 19, No. 11

VLDB 2025 **ParSEval: Plan-aware Test Database Generation for SQL Equivalence Evaluation**, *Chunyu Chen, Zhengjie Miao, Yong Zhang, and Jiannan Wang*.
Proceedings of the VLDB Endowment (PVLDB), Vol. 18, No. 11

KDD 2025 **Refining Labeling Functions with Limited Labeled Data**, *Chenjie Li, Amir Gilad, Boris Glavic, Zhengjie Miao, and Sudeepa Roy*.
In Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2

CHI 2024 **Human-LLM Collaborative Annotation Through Effective Verification of LLM Labels**, *Xinru Wang, Hannah Kim, Sajjadur Rahman, Kushan Mitra, Zhengjie Miao*, .
Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems, May 2024

ICDE 2024 **Fairness-Aware Data Preparation for Entity Matching**, *Nima Shahbazi, Jin Wang, Zhengjie Miao, and Nikita Bhutani*.
IEEE 40th International Conference on Data Engineering

SIGMOD 2024 **Watchog: A Light-weight Contrastive Learning based Framework for Column Annotation**, *Zhengjie Miao, Jin Wang*, .
Proc. ACM Manag. Data 1, 4, Article 272 (December 2023)

SIGMOD 2022 **Understanding Queries by Conditional Instances**, *Amir Gilad∗, Zhengjie Miao∗, Sudeepa Roy, and Jun Yang*, * denotes equal contribution, .
Proceedings of the 2022 ACM SIGMOD International Conference on Management of Data, June 2022

SIGMOD 2021 **Rotom: A Meta-Learned Data Augmentation Framework for Entity Matching, Data Cleaning, Text Classification, and Beyond**, *Zhengjie Miao, Yuliang Li, and Xiaolan Wang*, .
ACM SIGMOD International Conference on Management of Data, June 2021

SIGMOD 2021 **Putting Things into Context: Rich Explanations for Query Answers using Join Graphs**, *Chenjie Li, Zhengjie Miao, Qitian Zeng, Boris Glavic, and Sudeepa Roy*, .
ACM SIGMOD International Conference on Management of Data, June 2021

WWW 2020 **Snippext: Semi-supervised Opinion Mining with Augmented Data**, *Zhengjie Miao, Yuliang Li, Xiaolan Wang, and Wang-Chiew Tan*, .
The Web Conference 2020, April 2020

SIGMOD 2019 **Explaining Wrong Queries Using Small Examples**, *Zhengjie Miao, Sudeepa Roy, and Jun Yang*, .
ACM SIGMOD International Conference on Management of Data, June 2019

SIGMOD 2019 **Going Beyond Provenance: Explaining Query Answers with Pattern-based Counterbalances**, *Zhengjie Miao*, *Qitian Zeng*, *Boris Glavic, and Sudeepa Roy*, * denotes equal contribution, .
ACM SIGMOD International Conference on Management of Data, June 2019

CIDR 2017 **Combining Design and Performance in a Data Visualization Management System**, *Eugene Wu, Fotis Psallidas, Zhengjie Miao, Haoci Zhang, Laura Rettig, Yifan Wu, Thibault Sellam*, .
Conference on Innovative Data Systems Research, Jan 2017

Short/Demonstration Papers

VLDB 2026 **ParSEval: Interactive Counterexample-driven Evaluation for Text-to-SQL**, *Chunyu Chen, Zhengjie Miao, Yong Zhang, and Jiannan Wang*, .
Proceedings of the VLDB Endowment (PVLDB), Vol. 19, No. 12

DATAI 2025 **CleanAgent: Automating Data Standardization with LLM-based Agents**, *Danrui Qi, Zhengjie Miao, Jiannan Wang*, .
In 2nd International Workshop on Data-driven AI (DATAI), in conjunction with VLDB

ICDE 2025 **LineageX: A Column Lineage Extraction System for SQL**, *Shi Heng Zhang, Zhengjie Miao, Jiannan Wang*, .
In 2025 IEEE 41st International Conference on Data Engineering (ICDE) (pp. 4604-4607)

DataEd 2024 **What Teaching Databases Taught Us about Researching Databases: Extended Talk Abstract**, *Jun Yang, Amir Gilad, Yihao Hu, Hanze Meng, Zhengjie Miao, Sudeepa Roy, and Kristin Stephens-Martinez*, .
Proceedings of the 3rd International Workshop on Data Systems Education: Bridging education practice with education research, June 2024

ICDT 2024 **How Database Theory Helps Teach Relational Queries in Database Education (Invited Talk)**, *Sudeepa Roy, Amir Gilad, Yihao Hu, Hanze Meng, Zhengjie Miao, Kristin Stephens-Martinez, and Jun Yang*, .
27th International Conference on Database Theory

SIGMOD 2023 **Characterizing and Verifying Queries Via CInsGen**, *Hanze Meng, Zhengjie Miao, Amir Gilad, Sudeepa Roy, and Jun Yang*, .
Proceedings of the 2023 ACM SIGMOD International Conference on Management of Data, June 2023

SIGCSE 2022 **I-Rex: An Interactive Relational Query Debugger for SQL**, *Yihao Hu, Zhengjie Miao, Zhiming Leong, Haechan Lim, Zachary Zheng, Sudeepa Roy, Kristin Stephens-Martinez, and Jun Yang*.
The 53rd ACM Technical Symposium on Computer Science Education (SIGCSE), Volume 2

VLDB 2022 **CAJADE: Explaining query results by augmenting provenance with context**, *Chenjie Li, Juseung Lee, Zhengjie Miao, Boris Glavic, and Sudeepa Roy*, .
Proceedings of the VLDB Endowment, Vol. 15, No. 12

VLDB 2021 **Data Augmentation for ML-driven Data Preparation and Integration (Tutorial)**, *Yuliang Li, Xiaolan Wang, Zhengjie Miao, and Wang-Chiew Tan*, .
Proceedings of the VLDB Endowment (PVLDB), Vol. 14, No. 12

VLDB 2020 **I-Rex: An Interactive Relational Query Explainer for SQL**, *Zhengjie Miao, Tiangang Chen, Alexander Bendeck, Kevin Day, Sudeepa Roy, and Jun Yang*, .
Proceedings of the VLDB Endowment (PVLDB), Vol. 13, No. 12

VLDB 2019 **CAPE: Explaining Outliers by Counterbalancing**, *Zhengjie Miao*, Qitian Zeng*, Chenjie Li, Boris Glavic, Oliver Kennedy, and Sudeepa Roy*, * denotes equal contribution, .
Proceedings of the VLDB Endowment (PVLDB), Vol. 12, No. 12

VLDB 2019 **LensXPlain: Visualizing and Explaining Contributing Subsets for Aggregate Query Answers**, *Zhengjie Miao, Andrew Lee, and Sudeepa Roy*, .
Proceedings of the VLDB Endowment (PVLDB), Vol. 12, No. 12

SIGMOD 2019 **RATest: Explaining Wrong Relational Queries Using Small Examples**, *Zhengjie Miao, Sudeepa Roy, and Jun Yang*, .
ACM SIGMOD International Conference on Management of Data, June 2019

Mentoring

Summer 2025 Undergrad@SFU: Kunpeng Zhang, SFU VPR Undergraduate Student Research Award

Summer 2025 Undergrad@SFU: Eitan Barylko, NSERC Undergraduate Student Research Award

Fall 2024 - Ph.D. Student@SFU: Shi Heng Zhang, co-supervised with Jiannan Wang

Fall 2024 - Ph.D. Student@SFU: Chunyu Chen, co-supervised with Jiannan Wang

Fall 2024 - Ph.D. Student@SFU: Xinjie Hu (transferred from the M.S. program in Fall 2025)

Spring 2024 - Ph.D. Student@SFU: Xinyi Huang (transferred from the M.S. program in Fall 2025)

Summer 2023 Research Intern@Megagon Labs: Xinru Wang (Purdue University), co-mentored with Hannah Kim

Summer 2023 Research Intern@Megagon Labs: Nima Shahbazi (University of Illinois Chicago), co-mentored with Jin Wang, Nikita Bhutani

2020-2021 M.S. Project@Duke: Qiulin Li

2019-2020 M.S. Project@Duke: Tiangang Chen

2019-2020 M.S. Project@Duke: Kehan Lyu

2020 Duke Undergraduate CS Plus: Alexander Bendeck (now PhD student at Georgia Tech), Kevin Day, Jeffrey Luo

2020 Undergraduate Independent Study@Duke: Jeremy Cohen

Recent Talks

Test Database Generation for Text-to-SQL Evaluation and Beyond

- The Chinese University of Hong Kong, Department of CSE seminar, Aug 2025
- Northwest Database Society (NWDS) Annual Meeting, Feb 2025

- SFU CS Colloquium, Dec 2024

Watchog: A Light-weight Contrastive Learning based Framework for Column Annotation

- SIGMOD, Jun 2024
- MSR Asia-Pacific Vancouver Visit Day at SFU CS, May 2024

Understanding Queries by Examples

- Northeastern University, Nov 2023

Teaching

Fall 2026 **Instructor**, *CMPT 354: Database Systems I*, SFU.
Spring 2026 **Instructor**, *CMPT 733: Big Data Lab II*, SFU.
Fall 2025 **Instructor**, *CMPT 354: Database Systems I*, SFU.
Summer 2025 **Instructor**, *CMPT 354: Database Systems I*, SFU.
Spring 2025 **Instructor**, *CMPT 733: Big Data Lab II*, SFU.
Summer 2024 **Instructor**, *CMPT 354: Database Systems I*, SFU.
Spring 2024 **Instructor**, *CMPT 459/984: Special Topics in Database Systems*, SFU.
Spring 2019 **TA**, *CompSci 316: Introduction to Database Systems*, Duke University.
Spring 2018 **TA**, *CompSci 216: Everything Data*, Duke University.

Professional Services

Organizer

2026 Submission Co-chair, Workshop on Systems for Data-centric Agents with Human-in-the-loop (DASHSys)

Program Committee Member

2027 SIGMOD, ICDE
2026 Workshop on Human-In-the-Loop Data Analytics (HILDA)
2026 SIGMOD, VLDB, ICDE
2025 Workshop on Human-In-the-Loop Data Analytics (HILDA)
2025 SIGMOD
2025 ICDE (demonstration track)
2025 Workshop on Data-AI Systems (DAIS)
2024 Workshop on Human-In-the-Loop Data Analytics (HILDA)
2024 VLDB
2024 SIGMOD
2023 Workshop on Human-In-the-Loop Data Analytics (HILDA)
2023 ICDE
2023 ICDE (demonstration track)
2019 VLDB Reproducibility

Reviewer

2025 The VLDB Journal
2023 ACM Journal of Data and Information Quality
2020 ACM Transactions on Database Systems

External Reviewer

2022 ICDE
2021 ICDT